

ThaliMetrics

One tap at the tray return. Two problems solved.

The Official Challenge

"How might we develop an innovative, affordable, accessible, and sustainable solution that improves nutrition and overall health while reducing the environmental impact of food choices?" — Nutrition Week Hackathon 2026 Problem Statement

ThaliMetrics addresses three of the official focus areas directly: **Campus nutrition**, **Food waste & sustainability**, and **Micronutrient gaps**.

Problem Statement

THE PROBLEM

Hostel messes waste large amounts of food every day, while the students eating in those same messes commonly fall short of basic nutrients — and neither side can see the other's part of the story. NFHS-5 shows anaemia among young Indian women (15–19 yrs) at 59.1%, and ICMR-NIN sets protein RDA at 0.83 g/kg/day, a bar many students don't reliably clear.^{1,2} Hostel mess waste audits have recorded 3–6 kg of leftover food per meal slot in a single block.³ Existing nutrition apps also miss how Indian students actually eat — shared bowls, refills, second helpings — because they assume one static plate per meal.

Solution Statement

THE SOLUTION

ThaliMetrics is a tap-and-log app built around two realistic sessions per meal: **"I'm taking food"** (log each serving as you take it, including seconds) and **"I'm leaving / logging waste"** (show exactly how much of what you took is left, scaled to your own portion — never a generic 0–100%). The same data stream does two jobs: gives each student a weekly digest covering every tracked nutrient, and gives the mess a per-dish, per-day waste dashboard with locally-relevant swap suggestions.

Core Features

- 1. Session-based logging with refills**

Take food, add a second helping later, then log waste separately when leaving — matches real eating behaviour instead of a single static plate.
- 2. Vessel fill-sliders & roti quarter grid**

A translucent slider rises inside a bowl or glass shape for gravies, dry sabzis, and liquids; roti uses a 4-quarter tap grid. No typing, under five seconds per item.
- 3. Absolute-scale waste logging**

The waste slider for each dish is scaled to exactly what that student took — someone who took 3 bowls and someone who took 1 are never compared on the same 0–100% bar.
- 4. Mess Menu admin control**

The mess sets which dishes are actually on today's menu; students only see and log what's picked, matching the real daily menu plan.
- 5. Mess waste dashboard**

Today / This Week / This Month views, per-dish waste %, and a daily categorized breakdown — plus a swap suggestion for the worst-waste dish using locally available Tamil Nadu produce, aligned with FSSAI Eat Right Campus guidance.⁴
- 6. Full micronutrient weekly digest**

Protein, iron, calcium, vitamin A, vitamin C, folate, vitamin B12, zinc, and fiber — none skipped — against ICMR-NIN 2020 reference intakes, segmented by the student's own gender and body weight.

Who Uses It

Students

- Log servings + refills, then log waste on leaving
- Full weekly nutrient digest, gender + weight aware
- Generic nudge to campus health centre if a gap persists — never a diagnosis or named supplement

Mess In-Charge / Chefs

- Set today's menu from a dish library
- Daily/weekly/monthly waste by dish
- Local, seasonal swap suggestions

Parents (opt-in, future)

- Weekly summary email, not real-time tracking
- Privacy-first, student-controlled sharing

Grounding Data

Data Point	Source
59.1% of young Indian women (15–19 yrs) are anaemic	NFHS-5, National Family Health Survey

Protein RDA: 0.83 g/kg body weight/day (both sexes)	ICMR-NIN RDA-2020
Iron RDA: 19 mg/day (men), 29 mg/day (women)	ICMR-NIN RDA-2020
Hostel mess leftovers reached 3–6 kg per meal slot in audited cases	Hostel mess waste-audit study, Rajasthan University
17% of food wasted at the consumer level in India	UNEP/WRAP Food Waste Index Report, 2021
Eat Right Campus certifies on food safety, healthy diets, waste management, local food promotion	FSSAI, Eat Right India

Future Prospects

- Real food photography in place of the current CSS/SVG vessel shapes
- Apple Health / Samsung Health integration for automatic BMI and activity-based targets
- Basic exercise suggestions alongside dietary nudges
- Voice-to-text menu planning support for mess staff
- Live backend + vendor integration, piloted first in a single hostel block, working toward full FSSAI Eat Right Campus certification for VIT

Feasibility Note

The current build is a fully working browser app — no backend, no hardware, no IoT sensors. It runs on a live GitHub Pages link today. Every core feature above has been tested end-to-end (session logging, refills, absolute waste scaling, dashboard aggregation, and gender-segmented nutrient math all verified against expected values, not just eyeballed).